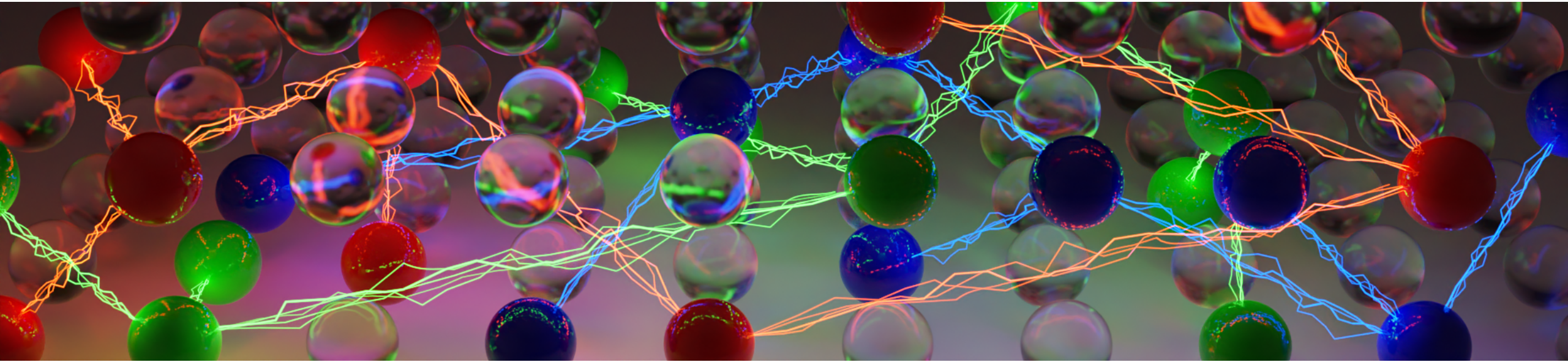




Practical SAT Solving

Lecture 6 – Preprocessing

Markus Iser, Dominik Schreiber | May 26, 2025

Conflict-driven Clause Learning (CDCL)

Recap: Last Time

- Efficient Unit Propagation
- Clause Forgetting
- Modern Decision Heuristics

Algorithm 1: CDCL(CNF Formula F , &Assignment $A \leftarrow \emptyset$)

```
1 if not PREPROCESSING then return UNSAT
2 while  $A$  is not complete do
3   UNIT PROPAGATION
4   if  $A$  falsifies a clause in  $F$  then
5     if decision level is 0 then return UNSAT
6     else
7       (clause, level)  $\leftarrow$  CONFLICT-ANALYSIS
8       add clause to  $F$  and backtrack to level
9       continue
10  if RESTART then backtrack to level 0
11  if CLEANUP then forget some learned clauses
12  BRANCHING
13 return SAT
```

Conflict-driven Clause Learning (CDCL)

Recap: Last Time

- Efficient Unit Propagation
- Clause Forgetting
- Modern Decision Heuristics

Today

- Preprocessing

Algorithm 2: CDCL(CNF Formula F , &Assignment $A \leftarrow \emptyset$)

```
1 if not PREPROCESSING then return UNSAT
2 while  $A$  is not complete do
3   UNIT PROPAGATION
4   if  $A$  falsifies a clause in  $F$  then
5     if decision level is 0 then return UNSAT
6     else
7       (clause, level)  $\leftarrow$  CONFLICT-ANALYSIS
8       add clause to  $F$  and backtrack to level
9       continue
10  if RESTART then backtrack to level 0
11  if CLEANUP then forget some learned clauses
12  BRANCHING
13 return SAT
```

Preprocessing

Preprocessing takes place between problem encoding and its solution.

Preprocessing is ...

- a form of **reencoding** a problem
- a form of **reasoning** itself

Classic Preprocessing Techniques

Conjecture: Smaller problems are easier to solve $\implies$ Try to reduce the size of the formula.

- Subsumption
- Self-subsuming Resolution
- (Bounded) Variable Elimination (BVE)

Subsumption

A clause C is subsumed by D iff $D \subseteq C$.

Subsumed clauses can be removed from the formula without changing satisfiability: $\forall D \subseteq C, D \models C$

Example

$\{a, b\}$ subsumes $\{a, b, c\}$ and $\{a, b, d\}$

Subsumption

A clause C is subsumed by D iff $D \subseteq C$.

Subsumed clauses can be removed from the formula without changing satisfiability: $\forall D \subseteq C, D \models C$

Example

$\{a, b\}$ subsumes $\{a, b, c\}$ and $\{a, b, d\}$

Implementation 1: Forward Subsumption

Select clause C and **check if it is subsumed** by any other clause $D \subseteq C$.

How to check if there exists a clause D that subsumes C ?

Subsumption

A clause C is subsumed by D iff $D \subseteq C$.

Subsumed clauses can be removed from the formula without changing satisfiability: $\forall D \subseteq C, D \models C$

Example

$\{a, b\}$ subsumes $\{a, b, c\}$ and $\{a, b, d\}$

Implementation 1: Forward Subsumption

Select clause C and **check if it is subsumed** by any other clause $D \subseteq C$.

Temporarily mark all literals in C as unsatisfied and propagate() to find subsuming clauses.

- **Optimization 1:** Use one-watched literal data-structure
- **Optimization 2:** Watch literals with the fewest occurrences
- **Optimization 3:** Keep literals sorted and perform merge-sort style subset check

Subsumption

Implementation 2: Backward Subsumption

Select clause D and **check if it subsumes** any other clause $C \supseteq D$.

Learned clauses can subsume other clauses.

Algorithm 3: Signature-based Subsumption

// Initialization:

1 **for** $clause \in formula$ **do**

2 $clause.signature = 0$

3 **for** $lit \in *clause$ **do**

4 $clause.signature |= 1ull \ll (id(lit)\%64)$

// Subsumption Check:

5 **if** $D.signature \& invert(C.signature) == 0$ **then**

 // Check if D subsumes C

Subsumption

Implementation 2: Backward Subsumption

Select clause D and **check if it subsumes** any other clause $C \supseteq D$.

Learned clauses can subsume other clauses.

How about the other way around?

Algorithm 4: Signature-based Subsumption

// Initialization:

1 **for** $clause \in formula$ **do**

2 $clause.signature = 0$

3 **for** $lit \in *clause$ **do**

4 $clause.signature |= 1ull \ll (id(lit)\%64)$

// Subsumption Check:

5 **if** $D.signature \& invert(C.signature) == 0$ **then**

 // Check if D subsumes C

Subsumption

Implementation 2: Backward Subsumption

Select clause D and **check if it subsumes** any other clause $C \supseteq D$.

Learned clauses can subsume other clauses.

- **Optimization 1:** Only check the clauses of the variable with the fewest occurrences (scales to large formulas, might miss some subsumptions)
- **Optimization 2:** Use signatures to skip the majority of subsumption checks (cf. Bloom filters)

Algorithm 5: Signature-based Subsumption

// Initialization:

1 **for** $clause \in formula$ **do**

2 $clause.signature = 0$

3 **for** $lit \in *clause$ **do**

4 $clause.signature |= 1ull \ll (id(lit)\%64)$

// Subsumption Check:

5 **if** $D.signature \& invert(C.signature) == 0$ **then**

 // Check if D subsumes C

Self-Subsuming Resolution

Applicable if the resolvent of C and another clause D subsumes C .

If $C \otimes_x D \subseteq C$ then C can be **replaced** by $C \otimes_x D$.

Example

Let $\otimes_f$ be the resolution operator on variable f .

$$C := \{\neg b, \neg e, f, \neg h\} \quad D := \{\neg b, \neg e, \neg f\} \quad E := C \otimes_f D = \{\neg b, \neg e, \neg h\}$$

→ Replace C by E (“clause strengthening”)

Self-Subsuming Resolution

Applicable if the resolvent of C and another clause D subsumes C .

If $C \otimes_x D \subseteq C$ then C can be **replaced** by $C \otimes_x D$.

Example

Let $\otimes_f$ be the resolution operator on variable f .

$$C := \{\neg b, \neg e, f, \neg h\} \quad D := \{\neg b, \neg e, \neg f\} \quad E := C \otimes_f D = \{\neg b, \neg e, \neg h\}$$

→ Replace C by E (“clause strengthening”)

Implementation

- **Integrate with subsumption:** Allow at most one literal of D to occur negated in C
- **Variant:** On-the-fly subsumption/strengthening of reason clauses during conflict analysis

Bounded Variable Elimination (BVE)

Let $S_x, S_{\bar{x}} \subseteq F$ be the sets of all clauses containing x resp. $\bar{x}$,
and let $R = \{C \otimes_x D \mid C \in S_x, D \in S_{\bar{x}}\}$ be the set of all resolvents on x .

The formulas F and $F' := (F \setminus (S_x \cup S_{\bar{x}})) \cup R$ are **equisatisfiable** but not equivalent.

Bounded Variable Elimination (BVE)

Let $S_x, S_{\bar{x}} \subseteq F$ be the sets of all clauses containing x resp. $\bar{x}$,
and let $R = \{C \otimes_x D \mid C \in S_x, D \in S_{\bar{x}}\}$ be the set of all resolvents on x .

The formulas F and $F' := (F \setminus (S_x \cup S_{\bar{x}})) \cup R$ are **equisatisfiable** but not equivalent.

Bounded Variable Elimination (BVE)

Eliminate variable only if the formula **size does not increase** (too much).

- **Note 1:** Variables of removed clauses can be rescheduled for further elimination attempts
- **Note 2:** Resolvent can trigger further subsumptions and vice versa
- **Variant:** Incrementally Relaxed BVE: Increase bound each round if formula size did not increase too much
- **Optimizations:** Perform check only for bounded clause size, resolvent size, or variable occurrence count

BVE is particularly effective in presence of functional definitions (cf. Tseitin encoding)

Blocked Clause Elimination (BCE)

A clause $\{x\} \cup C$ is blocked in F by x if either x is **pure** in F or for every clause $\{\neg x\} \cup D$ in F the resolvent $C \cup D$ is a **tautology**.

→ Dead ends in the resolution graph: no proof beyond this point.

Blocked clause elimination (BCE) has a unique fixpoint, and **preserves satisfiability**.

Example

$$F := (a \vee b) \wedge (a \vee \neg b \vee \neg c) \wedge (\neg a \vee c)$$

First clause is not blocked, second is blocked by both a and $\neg c$, third is blocked by c .

Blocked Clause Elimination (BCE)

A clause $\{x\} \cup C$ is blocked in F by x if either x is **pure** in F or for every clause $\{\neg x\} \cup D$ in F the resolvent $C \cup D$ is a **tautology**.

→ Dead ends in the resolution graph: no proof beyond this point.

Blocked clause elimination (BCE) has a unique fixpoint, and **preserves satisfiability**.

Example

$$F := (a \vee b) \wedge (a \vee \neg b \vee \neg c) \wedge (\neg a \vee c)$$

First clause is not blocked, second is blocked by both a and $\neg c$, third is blocked by c .

- Effectiveness of BVE can be increased by interleaving it with BCE
- Together with BVE: relationship with **circuit-level simplification techniques**

Blocked Clause Elimination (BCE)

A clause $\{x\} \cup C$ is blocked in F by x if either x is **pure** in F or for every clause $\{\neg x\} \cup D$ in F the resolvent $C \cup D$ is a **tautology**.

→ Dead ends in the resolution graph: no proof beyond this point.

Blocked clause elimination (BCE) has a unique fixpoint, and **preserves satisfiability**.

Example

$$F := (a \vee b) \wedge (a \vee \neg b \vee \neg c) \wedge (\neg a \vee c)$$

First clause is not blocked, second is blocked by both a and $\neg c$, third is blocked by c .

- Effectiveness of BVE can be increased by interleaving it with BCE
- Together with BVE: relationship with **circuit-level simplification techniques**

Generalization: Covered Clauses

A clause C is covered if it can be **turned into a blocked clause** by adding a covered literal.

A literal x is covered in C , if C contains a literal y such that all non-tautological resolvents of C on y contain x .

Solution Reconstruction

Many preprocessing techniques remove clauses or variables from a formula in a mere satisfiability-preserving way, such that the solution to the preprocessed formula needs some processing in order to be a solution to the original formula.

Reconstruction Algorithm

Keep track of eliminated variables (BVE) and clauses (BCE) in a solution reconstruction stack S , and if a model is found, use it to reconstruct a solution to the original formula.

The order is important, such that the last literal-clause pair (l, C) in S needs to be the first to be processed.

Algorithm 6: Solution Reconstruction

Data: Assignment A , Stack S

- 1 **while** S is not empty **do**
- 2 remove the last literal-clause pair (l, C) from S ;
- 3 **if** C is not satisfied by A **then**
- 4 $A := (A \setminus \{l = 0\}) \cup \{l = 1\}$
- 5 If variables remain unassigned in A , then assign them an arbitrary value.

Recap.

Classic Techniques

- Subsumption and Self-subsuming Resolution
- Bounded Variable Elimination
- Blocked Clause Elimination
- Solution Reconstruction

Next Up

Relationship between preprocessing techniques and gate encodings

Relationship with Gate Encodings

Tseitin encoding E of a gate with output o , function g , and input literals $x_1, \dots, x_n$:

$$E \equiv o \leftrightarrow g(x_1, \dots, x_n)$$

Properties of Gate Encodings

Let a Tseitin encoding $E \equiv o \leftrightarrow g(x_1, \dots, x_n)$ be given, and let $A(X) := \{T \cup \{\bar{x} \mid x \in X \setminus T\} \mid T \in 2^X\}$ denote the set of all assignments to variables in $X := \{x_1, \dots, x_n\}$.

For each input assignment $I \in A(\{x_1, \dots, x_n\})$,

1. there exists **at least one** output assignment $O \in \{o, \bar{o}\}$ such that $I \cup O \models E$ (left-totality)
 2. there exists **at most one** output assignment $O \in \{o, \bar{o}\}$ such that $I \cup O \models E$ (right-uniqueness)
- The output is uniquely determined by the input, such that either $I, o \models E$ and $I, \bar{o} \not\models E$ or vice versa.

Relationship with Gate Encodings

From the left-totality it follows that a Tseitin encoding E is a satisfiable set of blocked clauses.

Left-Totality of Gate Encodings

Let a Tseitin encoding $E \equiv o \leftrightarrow g(x_1, \dots, x_n)$ be given, it holds that

1. for each clause $C \in E$, either $o \in C$ or $\bar{o} \in C$

Proof: The existence of a clause $C \in E$ such that $o \notin \text{vars}(C)$ would contradict left-totality, because the assignment falsifying C , falsifies E for any assignment to o .

2. and all resolvents $R \in E_o \otimes_o E_{\bar{o}}$ are tautological.

Proof: The existence of a non-tautological resolvent $R \in E_o \otimes_o E_{\bar{o}}$ would contradict left-totality, because $E \models R$ and $o \notin \text{vars}(R)$, such that the assignment falsifying R , falsifies E for any assignment to o .

Relationship with Gate Encodings

From the left-totality it follows that a Tseitin encoding E is a satisfiable set of blocked clauses.

Example (Tseitin encoding $E \equiv o \leftrightarrow x \wedge y$)

Let a Tseitin encoding $E := \{\{\neg o, x\}, \{\neg o, y\}, \{o, \neg x, \neg y\}\} \equiv o \leftrightarrow x \wedge y$ be given, it holds that

1. all resolvents in $E_o \otimes_o E_{\bar{o}} = \{\{x, \neg x, \neg y\}, \{y, \neg x, \neg y\}\} \equiv \top$ are tautological,
2. and Blocked Clause Elimination (BCE) would remove all clauses from E .

Questions:

- What does BCE do to $F = \{\{o\}\} \cup E$?
- What does BCE do to $F = \{\{\neg o\}\} \cup E$?
- What does BCE do to $F = \{\{q\}, \{\neg q, o, p\}, \{\neg q, \neg o, \neg p\}\} \cup E$?

Relationship with Gate Encodings

Resolving the clauses of a gate encoding on the output literal o results in a set of tautological clauses.

Idea: Optimized Variable Elimination for Gate Encodings E

Let a formula $F = E \cup R$ with gate clauses E and remainder R be given.

Apply variable elimination as follows:

$$\begin{aligned}(E_x \cup R_x) \otimes (E_{\bar{x}} \cup R_{\bar{x}}) &\equiv (E_x \otimes R_{\bar{x}}) \cup (R_x \otimes E_{\bar{x}}) \cup (R_x \otimes R_{\bar{x}}) \cup (E_x \otimes E_{\bar{x}}) \\ &\equiv (E_x \otimes R_{\bar{x}}) \cup (R_x \otimes E_{\bar{x}}) \cup (R_x \otimes R_{\bar{x}}) && (E_x \otimes E_{\bar{x}} \equiv \top) \\ &\equiv (E_x \otimes R_{\bar{x}}) \cup (R_x \otimes E_{\bar{x}}) && ((E_x \otimes R_{\bar{x}}) \cup (R_x \otimes E_{\bar{x}}) \models R_x \otimes R_{\bar{x}})\end{aligned}$$

Proof Idea:

Each clause $c \in R_x \otimes R_{\bar{x}}$, derived by resolving $c_x \in R_x$ and $c_{\bar{x}} \in R_{\bar{x}}$, can also be derived by resolving clauses in $R_{\bar{x}} \otimes E_x$ and $E_{\bar{x}} \otimes R_x$.

Recap.

Recap.

- Classic Preprocessing Techniques:
Subsumption, Self-subsuming Resolution, Bounded Variable Elimination, Blocked Clause Elimination
- Relationship between Preprocessing Techniques and Gate Encodings

Next Up

Propagation-based Redundancy Notions

Propagation-based Redundancy Notions

Let a formula F , and a literal x be given.

Failed Literal Probing

If $F \wedge x \vdash_{UP} \perp$, then $F \models \neg x$

$\implies$ add $\{\neg x\}$ to F

Example (Failed Literal Probing)

Let $F := \{\{a, b\}, \{a, \neg b\}\}$.

Probing with $\neg a$ results in a conflict, i.e., $F \wedge \neg a \vdash_{UP} \perp$.

Ergo, we can deduce $F \equiv F \wedge a$.

Propagation-based Redundancy Notions

Let a formula F , a clause $C \in F$, and a literal $x \in C$ be given.

Asymmetric Literal Elimination (ALE)

If $F \setminus C \wedge \overline{C \setminus \{x\}} \vdash_{UP} \bar{x}$, then $F \models C \setminus \{x\}$.

$\implies$ strengthen C to $C \setminus \{x\}$

Example (Asymmetric Literal Elimination (ALE))

Let $F := \{\{a, b\}, \{\neg b, \neg c\}, \{a, c, d\}\}$, $C := \{a, c, d\}$, and $x := c$.

ALE results in the following propagation: $\{\{a, b\}, \{\neg b, \neg c\}, \{\neg a\}, \{\neg d\}\} \vdash_{UP} \neg c$.

Ergo, we can deduce $F \models \{a, d\}$.

F can not have a model which satisfies $\{a, c, d\}$, but not $\{a, d\}$.

Propagation-based Redundancy Notions

Let a formula F , a clause $C \in F$, and a literal $x \in C$ be given.

Asymmetric Tautology Elimination (ATE)

If $F \setminus C \wedge \overline{C} \vdash_{UP} \perp$, then $F \models C$.

$\implies$ remove C from F

Example (Asymmetric Tautology Elimination (ATE))

Let $F := \{\{a, b, c\}, \{\neg b, d\}, \{a, c, d\}\}$, and $C := \{a, c, d\}$.

ATE results in the following propagation: $\{\{a, b, c\}, \{\neg b, d\}, \{\neg a\}, \{\neg c\}, \{\neg d\}\} \vdash_{UP} \perp$,

Ergo, we can deduce $F \equiv F \setminus \{a, c, d\}$.

$\{a, c, d\}$ follows from the other clauses in F .

Propagation-based Redundancy Notions

Variants and Optimizations

- **Hidden Tautology Elimination (HTE) / Hidden Literal Elimination (HLE)**

Restricted forms of ATE/ALE which only propagate over binary clauses.

- **Distillation / Vivification**

Interleave assignment and propagation to detect ATs / ALs early on.

- **Avoidance of Redundant Propagations**

Sort literals and clauses in a formula to simulate a trie, and reuse propagations that share the same prefix.

Scheduling of Preprocessing Techniques

At a point where one technique is unable to make further progress, another technique might be applicable and even modify the problem in a way that the first technique can make further progress.

Scheduling of Preprocessing Techniques

- **Heuristic Bounds**

Bound the number of applications of a technique.

- **Scheduling of Techniques**

Non-trivial, benefit of techniques depends on the formula.

- **Interleaving of Techniques**

Apply techniques in a round-robin fashion.

- **Inprocessing**

Interleave search and preprocessing.

Recap.

Recap.

- Classic Preprocessing Techniques:
Subsumption, Self-subsuming Resolution, Bounded Variable Elimination, Blocked Clause Elimination
- Relationship between Preprocessing Techniques and Gate Encodings
- Propagation-based Redundancy Notions:
Failed Literal Probing, Asymmetric Literal Elimination, Asymmetric Tautology Elimination
- Variants and Optimizations of Redundancy Notions
- Scheduling of Preprocessing Techniques

Next Session

Redundancy Notions in the Context of Proof Systems